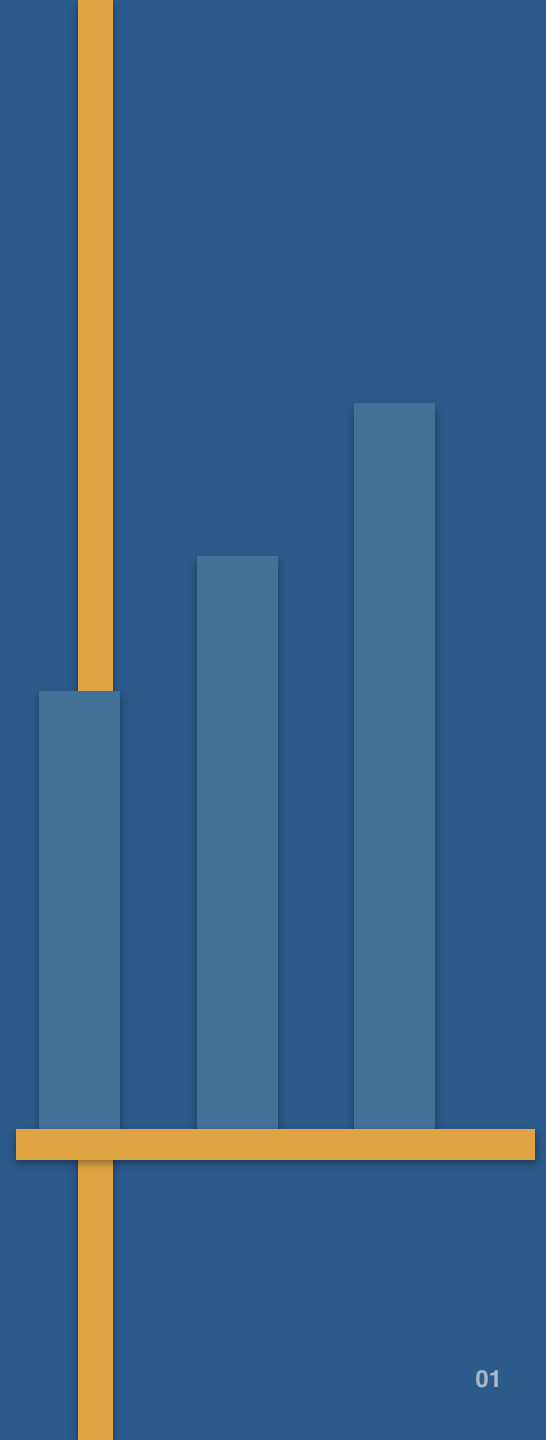
LOW-CODE APPLICATION DEVELOPMENT I ZOHO CREATOR

INTEGRATED GATEPASS & ACCESS MANAGEMENT SYSTEM

A digital workflow for visitor registration, visit approval and campus access monitoring

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Rajalakshmi Engineering College • November 2026



The gate is the first point of campus security

Manual registers make visitor information difficult to validate, retrieve and monitor.

MANUAL RECORDS

- Visitor and vehicle details are recorded separately
- Handwriting can be incomplete or inconsistent
- Past visits require manual searching

LIMITED VISIBILITY

- Host confirmation happens by phone
- Approval and visit schedules are not centralized
- Exit times are often missed

SECURITY CONCERNS

- No single view of visitors currently inside
- Personal details are visible in open registers
- Management lacks consolidated summaries

One digital record. A connected gate workflow.

Built on Zoho Creator with structured capture, Deluge automation, reports and integrated services.

01

REGISTER

Capture visitor, host, visit and vehicle details

02

VALIDATE

Check data, gate rules and vehicle constraints

03

APPROVE

Route selected visits to Security Officer

04

MONITOR

Check-in, check-out, reports and dashboard

Zoho Creator database · 11 Deluge form workflows · Zoho Mail · Zoho Calendar REST API

Four roles, clearly separated responsibilities

The Security Officer administers the system; front-desk, gate and host users work within their access.

SECURITY OFFICER

Approves or rejects visits; manages users and roles; views all records, reports and dashboard.

RECEPTIONIST

Registers visitors and vehicles; reviews reports, visitor board and dashboard.

SECURITY GUARD

Registers walk-ins; checks visitors in and out; monitors visitors currently inside.

HOST FACULTY / STAFF

Registers expected visitors; receives notifications; views own visits without access to visitor ID data.

A single Visitor Entry form

The central record captures one visit, its host and an optional vehicle.

VISIT DETAILS	Visit date • visitor type • purpose • entry gate • department
HOST DETAILS	Host name • host email
VISITOR DETAILS	Name • phone • ID proof type and number • number of persons
VEHICLE DETAILS	Vehicle present • type • registration number • parking area • material carried
SYSTEM FIELDS	Gate pass number • status • check-in/out timestamps • duration • parking fee

Fixed lists and conditional fields keep data entry consistent and reduce unnecessary input.

From visitor arrival to a completed visit

Selected categories trigger approval; security staff record the actual campus movement.

FORM OPEN
Defaults and user details

SUBMIT
Validate visitor and vehicle

ROUTE
Registered or Pending Approval

OFFICER
Approve / reject with remarks

GATE
Check-in, then check-out

CLOSE
Duration and parking fee

APPROVAL TRIGGERS

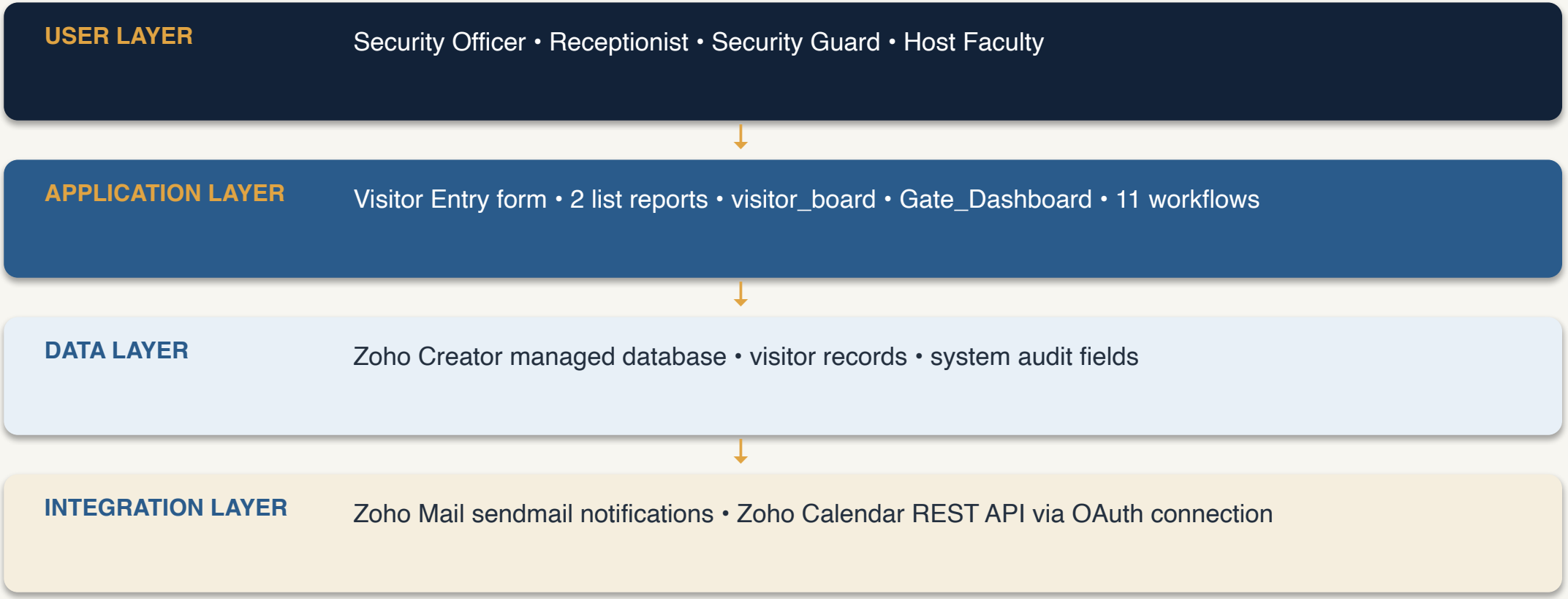
Contractor, official, truck and visits with 5+ persons are routed to Pending Approval.

GATE CONTROLS

Trucks use the Service Gate; duplicate active vehicle entries are blocked.

A low-code application with connected services

Four logical layers separate users, application logic, managed data and integrations.



11 Deluge workflows power the business rules

All eleven are form workflows on Visitor Entry; no standalone functions, schedules, blueprints or batch workflows.

ON LOAD

Defaults, visibility and access

ON EDIT LOAD

Restore defaults and access rules

VISITOR TYPE

Suggest purpose

VEHICLE TYPE

Assign parking area and rate

VEHICLE NUMBER

Normalize and validate registration

FORM VALIDATE

Enforce required fields and gate rules

ON SUCCESS

Generate pass and send registration email

STATUS CHANGE

Role checks and approval/rejection logic

CHECK-IN

Record entry timestamp

CHECK-OUT

Record exit, duration and fee

ON APPROVAL

Notify host and create Calendar event

Reports turn entries into operational visibility

Two list reports, one Kanban board and six dashboard charts.

VISITOR ENTRY REPORT

Search and review all submitted visit records.

VISITORS CURRENTLY INSIDE

Focus on active visits that have not checked out.

VISITOR BOARD

Kanban grouping by visit status.

01 Visitors by type

02 Vehicle type split

03 Daily visitor trend

04 Visitors by gate

05 Parking fee by vehicle type

06 Visitors by department

Notifications, scheduling and controlled visibility

Zoho services connect the gate workflow to hosts and the Security Officer.

ZOHO MAIL

- Host receives registration email with gate pass details
- Security Officer receives pending-approval alert
- Approval / rejection and check-out notifications

ZOHO CALENDAR

- OAuth connection: zoho_calendar_connection
- Approved visits are sent through the Calendar REST API
- Calendar event keeps approved visit details accessible

ROLE-BASED PERMISSIONS

Four roles with role-specific actions and visibility.

PERSONAL DATA PROTECTION

Host Faculty profile cannot access visitor ID data; record criteria and server-side checks support access control.

Testing covered valid paths and security rules

The report records 18 test cases and seven defects identified and closed.

18	7	11	6
TEST CASES	DEFECTS FOUND & CLOSED	FORM WORKFLOWS	DASHBOARD CHARTS

VALIDATION SCENARIOS

Phone and ID formats • visit date • persons count • vehicle number • truck gate • duplicate active vehicle.

WORKFLOW SCENARIOS

Pending approval • role-restricted approval • email notifications • Calendar event • check-in/out and fee calculation.

A visit moves through one connected journey

The report's example follows a parent visit registered at the Hostel Gate.

01 REGISTER	Receptionist records parent visit and host details.
02 NOTIFY	Host receives registration email; Security Officer is alerted when approval is required.
03 APPROVE	Security Officer approves or rejects with remarks.
04 SCHEDULE	Approved visit triggers host email and Zoho Calendar event.
05 COMPLETE	Security Guard checks the visitor in and out; duration and parking fee are captured.

Published across Stage and Production

Version 1.0 deployment is recorded as 09 November 2026.

DEVELOPMENT

Application configured in Zoho Creator trial edition.

STAGE

Version 1.0 promoted for staging and verification.

PRODUCTION

Version 1.0 published for operational access.

Deployment date • 09 November 2026

CONCLUSION

A more structured way to manage who enters campus and when.

The Integrated Gatepass and Access Management System brings visitor registration, validation, approval, gate movement, notifications and monitoring into one Zoho Creator application.

THANK YOU

Questions & Discussion